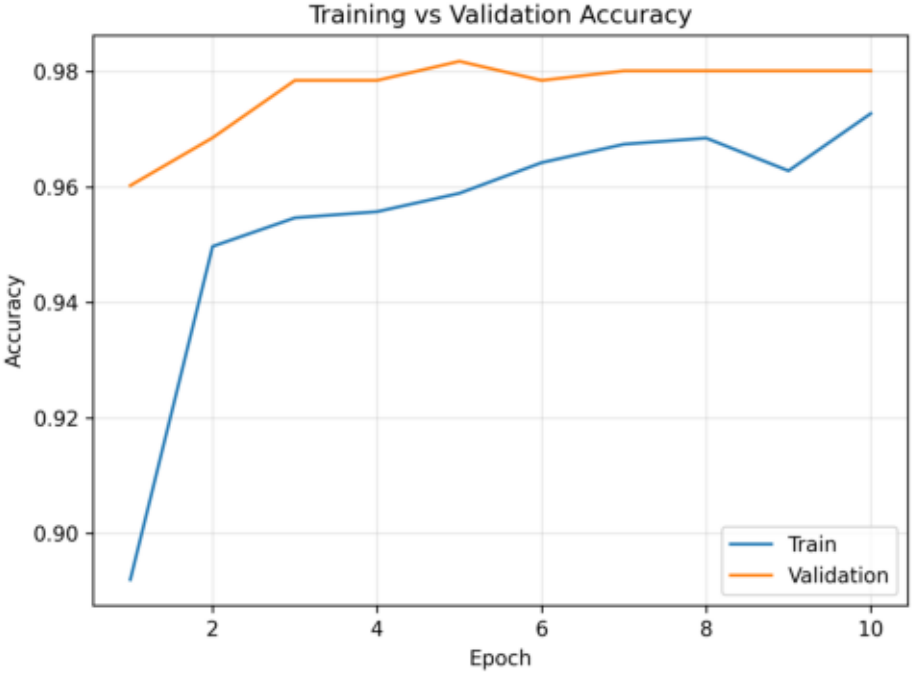
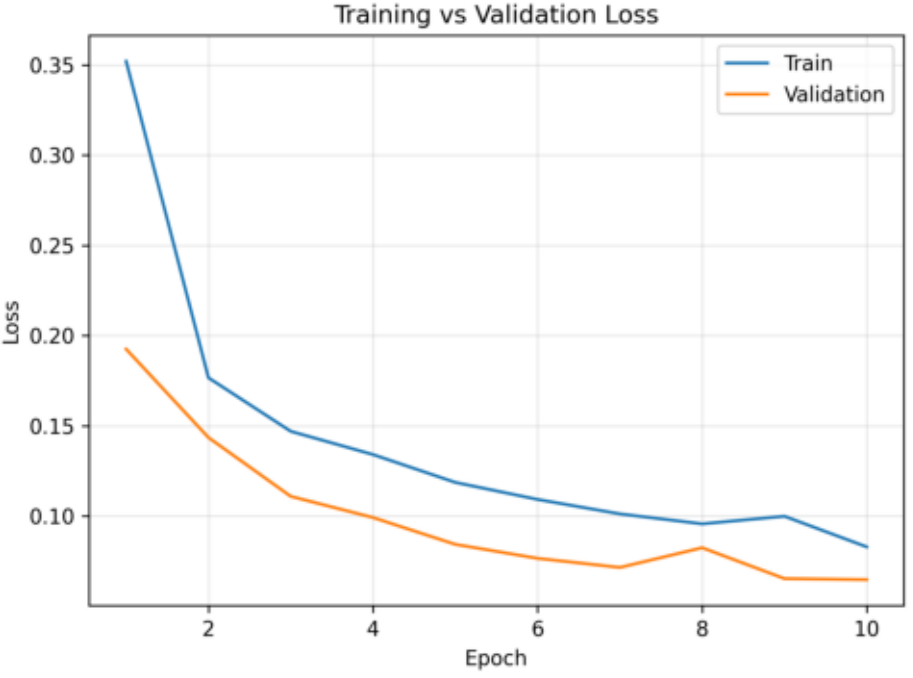
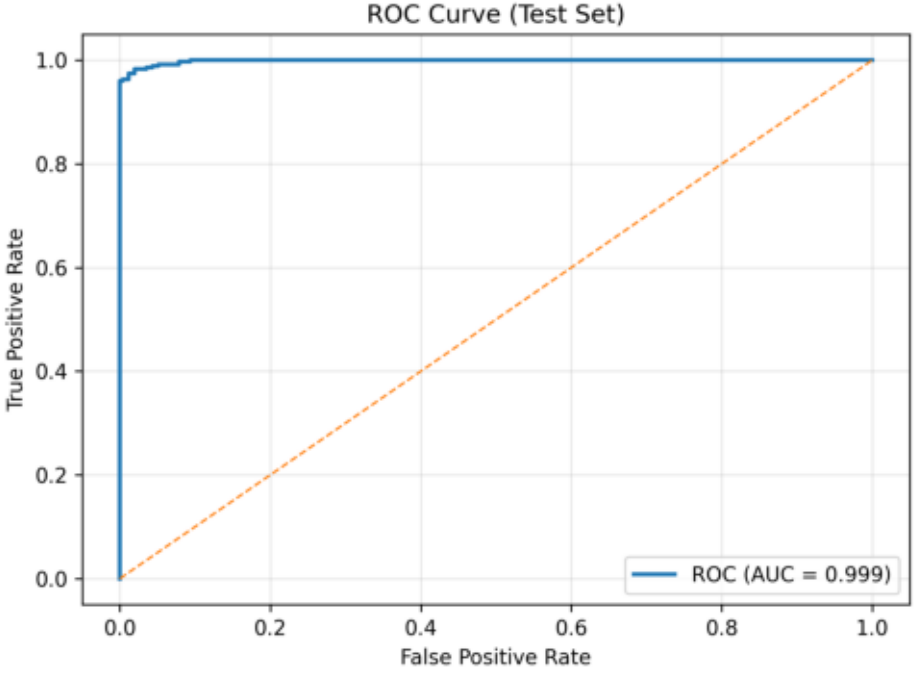
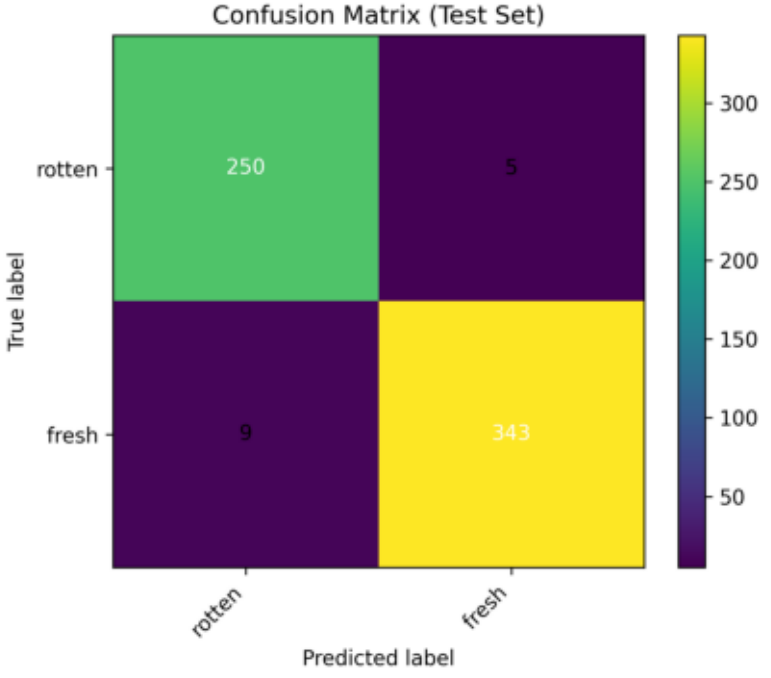
Apple Freshness Classification — Results Summary

```
Summary Metrics (Test Set):
EfficientNet-B0 (frozen backbone) | Test | Acc=0.9769  Prec=0.9856
```


Training vs Validation Accuracy
Accuracy
Epoch
0.98
0.96
0.94
0.92
0.90
2
4
6
8
10
Train
Validation


Training vs Validation Loss
Train
Validation
0.35
0.30
0.25
0.20
0.15
0.10
Loss
2
4
6
8
10
Epoch


ROC Curve (Test Set)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC (AUC = 0.999)


Confusion Matrix (Test Set)
rotten
fresh
True label
250
5
9
343
rotten
fresh
Predicted label
300
250
200
150
100
50

Comparison Table

Model	Split	Accuracy	Precision	Recall	F1	AUC
EfficientNet-B0 (frozen backbone)	Test	0.9769	0.9856	0.9744	0.98	0.9986